Clinical Care-Plan Case Study: Acute Decompensated Heart Failure with End-Stage Renal Disease

ADN Med-Surg II | Individual take-home clinical care-plan case study | Open book

Your patient

You are the primary nurse for the patient in room 418B on the 7:00 to 15:00 shift on Wednesday, September 2, 2026 (hospital day 3). Open the patient in the student EHR (Student View, or the patient file from your instructor). Everything you need is in the chart; set the simulation clock to 0700 and work the way you would at the start of a real shift.

How this case study works

This packet follows the nursing process: Assessment, Diagnosis, Planning, Implementation and Evaluation. You will also explain the pathophysiology and pharmacology behind your patient's care. Expect to spend about 6 hours. It is a take-home, individual, open-book assignment: use your textbooks, drug guide, notes and the chart.

Questions ask you to explain WHY, not only WHAT. Short, specific answers that use this patient's data earn more points than general textbook answers.

Part	Nursing process step	Estimated time
1	Assessment: Patient Data and Chart Review (Assessment)	75 min
2	Pathophysiology: The Disease Process (Assessment / Analysis of cues)	75 min
3	Pharmacology and Medication Safety (Assessment / Analysis (and safe implementation))	60 min
4	Nursing Diagnoses and Priorities (Diagnosis)	30 min
5	Planning and Implementation (Planning / Implementation)	60 min
6	Evaluation: What Happened and What Next? (Evaluation)	30 min
7	Clinical Judgment (Next-Generation NCLEX Style) (Evaluation / Clinical judgment)	30 min
	Estimated total	**6 h 0 min**

Scoring

- The packet is worth 100 points. A score of 78 or higher is passing.
- Each numbered item shows its point value. Points are given only for what is written in the box.

Writing so your pages can be scored from photos

- Write in dark pen or very dark pencil, print clearly, and stay inside the box. Write the answer in the box with the same number as the question.
- Write your name and date at the top of every page (the header prints a line for it).
- If you need more room, write "continued on the back" and put the item number on the back. Do not squeeze writing between boxes.
- Do not use abbreviations that are not on the approved list. Spell out drug names.
- When you finish: photograph EVERY page, one page per photo, on a flat surface in good light with all four corners and the page number in view. Check that every photo is readable before you upload.

Part 6 is sealed

- Do not read the sealed page at the start of Part 6 until Parts 1 to 5 are finished. Your plan must be based on what you knew at 0700.

Honor statement: I completed this work myself, using only my textbooks, notes, drug references and the patient chart.

Signature: _____ Date: _____

PART 1: ASSESSMENT: PATIENT DATA AND CHART REVIEW

Nursing process step: Assessment | About 75 minutes | 26 points

Collect and organize the data you would have at 0700. Where the chart does not document something, write "not documented." Use R and L for laterality.

1.1 Basic Patient Information *(4 pts)*

Where to look in the EHR: Patient Summary and Storyboard

Room Number:		**Age:**	
Gender:		**Admission Date:**	
Allergies (and reaction):		**Chief Complaint:**	
Primary Diagnosis:		**Code Status:**	
Fall Risk:			

1.2 Medical Diagnoses and How They Affect the Primary Diagnosis *(4 pts)*

List FIVE other medical diagnoses from this patient's chart. For each, explain how it affects her acute decompensated heart failure.

Where to look in the EHR: Chart Review > H&P; Encounters > Hospital problem list

Medical diagnosis	How it affects the primary diagnosis (acute decompensated heart failure)

1.3 Patient History *(3 pts)*

Where to look in the EHR: H&P; Case Management; Nursing Admission Note

Medical History:

Surgical History:

Family History:

Social History:

Home Situation and Discharge Needs:

Pre-Hospital Story of Patient:

1.4 Head-to-Toe Assessment at 0700 (use R and L for laterality) *(4 pts)*

Document the findings from the chart as of the start of your shift. If the chart does not document a finding, write "not documented."

Where to look in the EHR: Flowsheets; Nursing Progress Note 0645; Hospitalist Progress Note

Neurological:

Cardiovascular: Radial pulses R___ L___ Pedal pulses R___ L___ Cap refill _____

Respiratory: RUL ____ LUL ____ RLL ____ LLL ____ (breath sounds, effort, O2)

Integumentary:

Musculoskeletal:

Gastrointestinal:

Genitourinary:

IV / Lines / Devices Assessment:

Psychosocial / Cultural Assessment:

Erikson Stage:

1.5 Vital Sign Trend *(2 pts)*

Copy the vital signs at each time. Circle every abnormal value. Then, in the last box, describe the trend in one sentence.

Where to look in the EHR: Flowsheets > Vital Signs (or the Vitals graph)

Vital sign	08/31 0206 (ED arrival)	08/31 2000	09/01 2000	09/02 0400
Temperature				
Heart rate				
Blood pressure				
Respiratory rate				
SpO2 (%)				
Pain (0-10)				

Trend in one sentence:

1.6 Laboratory Data *(4 pts)*

Choose the 8 laboratory results most important for this patient. For each give the current value, the trend, and the normal range, and explain why it matters for THIS patient.

Where to look in the EHR: Lab Results (all columns; use Newest left/right as you like)

Lab	Results (current / trend / normal)	Why is this lab important for this patient?

1.7 Diagnostics *(2 pts)*

Where to look in the EHR: Chart Review > Imaging, Cardiology

Chest X-ray: location and results

Echocardiogram results:

EKG results:

Urinalysis results:

1.8 Chart Reconciliation *(3 pts)*

Real charts contain conflicts and gaps. Find THREE places where this chart is duplicated, conflicting, or needs clarification before you act. Say where you found each and who you would clarify it with.

Where to look in the EHR: Orders, Chart Review > Notes, Case Management

What is wrong or unclear (and where)	Who you would clarify with and what you would ask

Write one answer per numbered box, inside the box, in dark pen, neatly. Open book, individual work.

PART 2: PATHOPHYSIOLOGY: THE DISEASE PROCESS

Nursing process step: Assessment / Analysis of cues | About 75 minutes | 20 points

Explain WHY this is happening. Use your own words (plain language) and tie each idea to this patient.

2.1 Pathophysiology of Acute Decompensated Heart Failure (HFrEF) *(6 pts)*

In plain language and your own words, explain the pathophysiology of this patient's primary diagnosis: what is wrong with the heart, what the body does in response, and why that produces fluid in the lungs and legs.

2.2 Cause-and-Effect Chain *(3 pts)*

Fill in each box (one or two lines) to show how events in this patient lead from the trigger to her symptoms.

1. Trigger(s) in THIS patient	
▼	
2. What happens to blood volume and preload?	
▼	
3. What happens to the heart (EF, contractility)?	
▼	
4. Compensation by hormones/nerves, and its downside	

Write one answer per numbered box, inside the box, in dark pen, neatly. Open book, individual work.

▼

5. Backward effects (lungs)	

▼

6. Forward and systemic effects (kidneys, legs, liver)	

2.3 How Do You Know? Findings and Mechanisms *(4 pts)*

List EIGHT findings from this patient's assessment, labs or diagnostics that support acute decompensated heart failure. Beside each, explain the mechanism that causes it.

Where to look in the EHR: Notes, Flowsheets, Lab Results, Imaging

Finding in THIS patient (with value)	Why it happens (mechanism)

2.4a ESRD and fluid *(1 pt)*

Why is this patient's fluid overload harder to treat than a patient with normal kidneys? How is the excess fluid actually removed in her case, and when will a diuretic such as furosemide help or not help?

2.4b AFib and COPD *(1 pt)*

Explain how atrial fibrillation and COPD each can worsen her heart failure and her breathing.

2.4c Alcohol and tobacco *(1 pt)*

How may her heavy daily alcohol use and 1+ pack/day smoking have contributed to her illness, and what new risk does the alcohol create this week?

2.5 Treatment *(2 pts)*

What are we doing to treat this diagnosis? Include medical treatments AND nursing monitoring.

2.6 Potassium *(2 pts)*

Her potassium was 4.5 yesterday and is 5.0 this morning. Why does this matter in a patient with heart failure, atrial fibrillation and ESRD? Name two contributors for this patient, two signs you would look for, and your action.

Where to look in the EHR: Lab Results > Potassium

PART 3: PHARMACOLOGY AND MEDICATION SAFETY

Nursing process step: Assessment / Analysis (and safe implementation) | About 60 minutes | 14 points

Use your drug guide and the MAR (and the drug guide link inside the EHR).

3.1 Medication List *(6 pts)*

From the MAR, choose the EIGHT medications that matter most to you before the 0900 medication pass. List brand and generic names.

Where to look in the EHR: MAR (click the medication name for the drug guide)

Medication (brand and generic)	Drug class and mechanism (10 words or fewer)	Why is THIS patient taking it?	Key nursing check / hold parameter

3.2a Carvedilol *(2 pts)*

Carvedilol is a non-selective beta blocker and she has COPD. Why is it still ordered? What do you assess before giving it and when do you hold it? When in the day will you give it, given her dialysis?

3.2b Apixaban *(2 pts)*

Apixaban is a high-alert drug. List the risks specific to THIS patient, what you assess, and what you teach.

3.2c Lorazepam by CIWA-Ar *(2 pts)*

She has heavy daily alcohol use, dementia, and a PRN lorazepam order for CIWA-Ar of 10 or more (her last score was 6). What must you do before giving it, and what are the dangers for this patient?

3.3 Math From the Chart *(2 pts)*

Show your work in the box. Include units.

Where to look in the EHR: Flowsheets I&O and weights; Dialysis note

a) Her fluid restriction is 1,000 mL/day. Day shift (0700-1500) may receive 50% of the daily limit. How many mL may she have on your shift?

b) The hospitalist recorded 24-hour intake of 1,246 mL and output of 332 mL. What is the net balance?

c) Admission weight was 70 kg and the 09/01 2000 weight was 67.9 kg. How many pounds has she lost? (1 kg = 2.2 lb)

d) Dialysis removed 2,013 mL over 3.5 hours. What was the average ultrafiltration rate in mL/hour?

PART 4: NURSING DIAGNOSES AND PRIORITIES

Nursing process step: Diagnosis | About 30 minutes | 8 points

Use NANDA-I diagnoses. Write each as problem, "related to" (cause you can treat), and "as evidenced by" (specific data). Do not use a medical diagnosis as the "related to."

4.1 Prioritized Nursing Diagnoses *(6 pts)*

Write FOUR nursing diagnoses in priority order (1 = highest).

#	NANDA-I nursing diagnosis	Related to (etiology)	As evidenced by (at least 2 specific chart findings)

4.2 Why This Order? *(2 pts)*

Explain why your #1 diagnosis is first (use ABCs, Maslow, or acuity). Name one risk diagnosis you ranked lower that could become the top priority and what finding would move it up.

PART 5: PLANNING AND IMPLEMENTATION

Nursing process step: Planning / Implementation | About 60 minutes | 14 points

Plan the care for the 0700-1500 shift. SMART goal = Specific, Measurable, Achievable, Relevant, Time-bound, written about the PATIENT (not the nurse).

5.1a Plan for Priority Diagnosis #1: _____ *(2.5 pts)*

I = independent, C = collaborative, D = delegated.

SMART goal for this shift	Nursing interventions (at least 4: what, how often)	Rationale	I / C / D

5.1b Plan for Priority Diagnosis #2: _____ *(2.5 pts)*

SMART goal for this shift	Nursing interventions (at least 4: what, how often)	Rationale	I / C / D

5.2 Time-Sequenced Plan for the Shift *(3 pts)*

For each hour, list your priority actions IN ORDER. Include medications, hemodialysis, therapy, assessments, teaching, and communication, and say why that time.

Where to look in the EHR: Brain/Worklist, MAR, Orders, Secure Chat, Notes

Time	Priority actions in order	Why at this time
0700		
0800		
0900		
1000		
1100		
1200		
1300		
1400		

5.3 Calling the Provider (SBAR) *(2 pts)*

At 0715 you review the morning labs: potassium 5.0 (4.5 yesterday). She has dialysis scheduled at 0800, atrial fibrillation on telemetry, and is asymptomatic. You decide to call the hospitalist. Write what you would say.

Where to look in the EHR: Lab Results; ECG; Orders

S: Situation

Write one answer per numbered box, inside the box, in dark pen, neatly. Open book, individual work.

B: Background

A: Assessment

R: Recommendation / request

5.4 Delegation *(2 pts)*

Mark the LOWEST level of staff who may be assigned each task for this patient.

	RN only	LPN/LVN	UAP/PCT
Obtain the daily standing weight at 0600 and record it	☐	☐	☐
Listen to heart and lung sounds and interpret an S3 gallop and crackles	☐	☐	☐
Decide whether to give PRN lorazepam based on a new CIWA-Ar assessment	☐	☐	☐
Give the scheduled oral carvedilol after the RN confirms HR and BP parameters	☐	☐	☐

5.5 Teaching Plan *(2 pts)*

List FOUR teaching topics this patient needs before discharge. Adapt each to her barriers (dementia, lives alone, tobacco and alcohol use) and say how you will check understanding.

Teaching topic	How you will teach it (adapted to barriers)	How you will confirm learning

SEALED PAGE. Read only after Parts 1 to 5 are complete.

Chart entries recorded after 0700 (copied from the patient record):

09/02 08:12 Respiratory Therapy Note

Assessment
- O2 therapy: 2 L nasal cannula; SpO2 92%; RR 20.
- Breath sounds: Faint bibasilar crackles. Work of breathing: unlabored.
- Cough: Dry, worse lying flat.

Treatments
- Scheduled nebulizer: ipratropium-albuterol nebulizer solution 3 mL Every 6 hours; tolerated, no adverse effects.

09/02 08:30 Hospitalist Progress Note

Hospital Day 3
Interval events / subjective
Feels much better. Walking in hall without dyspnea; sleeping flat. Appetite good.
- Hemodialysis (3.5 h) (09/02/26 0800): completed

Objective
- Vitals (24 h): Tmax 98.4 F; HR 95-100; BP 158-168/88-92; RR 19-21; SpO2 90-92%; pain 0/10

09/02 11:42 Hemodialysis Treatment Note

Treatment
- Hemodialysis 0800-1130 (3.5 hours) via left upper arm AV fistula; two 15-gauge needles, good flows.
- Pre: BP 163/88, HR 95; weight above dry weight.
- Ultrafiltration goal and removed: 2013 mL. Heparin per protocol.
- Tolerated well without cramping or hypotension; needles removed, hemostasis achieved in 10 minutes, fistula thrill present.

Post-treatment
- Post-dialysis weight recorded. Give renal vitamin and hold antihypertensives if SBP below 100.

09/02 13:00 Physical Therapy Treatment Note

Session
- Therapeutic exercise, gait and transfer training; supervision / standby assist for transfers.
- Gait 150 feet without device; vitals stable with activity (T 98.2 F, HR 95, BP 163/88, RR 20, SpO2 92% on 2 l nasal cannula, pain 0/10).

Progress / plan
- Progressing toward discharge goals; continue daily PT.

1415 Late-shift update from the charge nurse (instructor addendum, not in the EHR)

- The patient's daughter called. She says her mother "has not been taking the water pill at home because it makes her get up all night," and she wants her mother home on Friday.
- When you ask her mother to repeat back the daily-weight rules, she remembers only "weigh every day." She cannot say when to call, and asks, "Can I have my ginger ale, I'm so thirsty?" There is a 355 mL can on her tray.
- The 1300 vital signs in the PT note show BP 163/88. Your 0900 note recorded that carvedilol and amlodipine were given.

PART 6: EVALUATION: WHAT HAPPENED AND WHAT NEXT?

Nursing process step: Evaluation | About 30 minutes | 8 points

Open the sealed page. It shows what the chart recorded during your shift, plus a late-shift update from your charge nurse. Do not change your Part 1-5 answers.

6.1 Evaluating Your Goals *(4 pts)*

For each goal you wrote in 5.1a and 5.1b, use data from the sealed page to say whether it was met, partly met, or not met, and what you will change.

Goal (copy from Part 5)	Data from the sealed page that you used	Met / Partly / Not	Revised plan

6.2 New Problems and Priority Actions *(2 pts)*

From the sealed page, name TWO problems that are still not resolved or are new, and the priority nursing action for each (including who you would notify).

6.3 Focused Nursing Note *(2 pts)*

Write a focused nursing progress note for the end of the shift (DAR, SOAP, or your school's format): data, action, and response, with the plan for the next shift.

PART 7: CLINICAL JUDGMENT (NEXT-GENERATION NCLEX STYLE)

Nursing process step: Evaluation / Clinical judgment | About 30 minutes | 10 points

Answer by marking the boxes. These items use the same patient. Some scenarios are hypothetical and say so.

7.1 Select All That Apply *(2 pts)*

Which teaching points are correct for this patient with heart failure and ESRD?

Select ALL that apply.

- ☐ Weigh yourself every morning on the same scale and write it down.
- ☐ Use a salt substitute (potassium chloride) to flavor food.
- ☐ Limit fluids as ordered (1,000 mL/day), including ice and soup.
- ☐ Take ibuprofen for aches because it is sold over the counter.
- ☐ Call if weight rises more than 2-3 pounds in a day or 5 pounds in a week.
- ☐ Skip a dialysis session if you feel good and have no swelling.
- ☐ Never miss a dialysis session, and ask before taking any new medicine.
- ☐ Drink extra water when thirsty to protect the kidneys.

7.2 Trend: Better, Same, or Worse? *(2 pts)*

Compare the patient at 0700 today with her ED arrival. Mark one box per row.

	Better	No real change	Worse
Oxygenation (SpO2 86% on arrival, 92% on 2 L now)	☐	☐	☐
Leg edema (2+ on arrival, 1+ now)	☐	☐	☐
Potassium (4.5 yesterday, 5.0 today)	☐	☐	☐
Hemoglobin (8.8 on arrival, 9.1 now)	☐	☐	☐

7.3 Bowtie *(2 pts)*

HYPOTHETICAL for this item: thirty minutes after her dialysis finishes she says "I feel faint." BP 88/50, HR 118 (atrial fibrillation), SpO2 91% on 2 L, skin pale and cool, lungs clearer than this morning.

Actions to Take (select 2)	Condition the patient is most likely experiencing (select 1)	Parameters to Monitor (select 2)

☐ Lay her flat with legs elevated and notify the provider ☐ Give the scheduled carvedilol and amlodipine now ☐ Give her furosemide ☐ Place the blood pressure cuff on her left arm ☐ Hold the scheduled carvedilol and amlodipine and notify the provider	☐ Hypotension from fluid removal (post-dialysis hypovolemia) with antihypertensives ☐ Acute pulmonary edema ☐ Hyperkalemia ☐ Alcohol withdrawal	☐ Blood pressure and heart rate/rhythm (telemetry) ☐ Level of consciousness ☐ Temperature ☐ Bowel sounds ☐ Pain score

7.4 Recognize Cues *(2 pts)*

Excerpt from a HYPOTHETICAL 0700 shift report. Which statements need follow-up NOW?

Draw a line under each statement that needs follow-up now. Then write the item numbers in the box below.

(1) Rate-controlled AFib on telemetry, HR 95, BP 163/88.

(2) Potassium 5.0 this morning (4.5 yesterday); dialysis at 0800.

(3) She slept flat for the first time since she came in.

(4) She drank 600 mL of ice water from a pitcher before 0700 and asked for more.

(5) CIWA-Ar was 6 at 0400, no tremor.

(6) Her daughter says she "seems more confused than usual" this morning.

(7) Left arm fistula: thrill present, no redness or swelling.

(8) The bed alarm sounded at 0630; she was trying to get up to the bathroom without calling.

Item numbers that need follow-up:

7.5 Priority Actions *(2 pts)*

HYPOTHETICAL: at 1000 she has SpO2 84% on 2 L, RR 30, pink frothy sputum, and crackles throughout. Number the actions in the order you would do them.

Number the actions 1 (first) to 5 (last).

[___] Call the rapid response team or provider with SBAR.

[___] Raise to high-Fowler's with legs dependent, stay with the patient.

[___] Apply a non-rebreather mask or increase oxygen per protocol.

[___] Anticipate and prepare orders (IV diuretic/urgent dialysis, ECG, labs).

[___] Reassess vital signs and document.